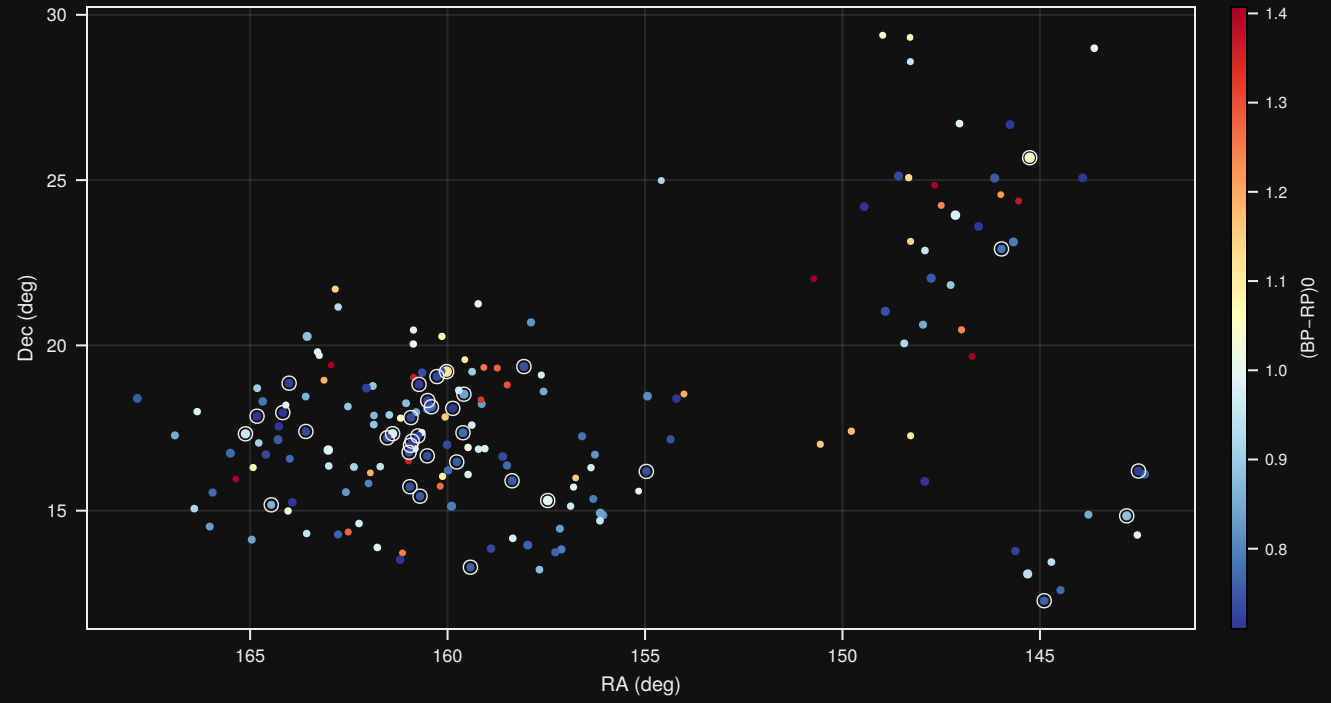
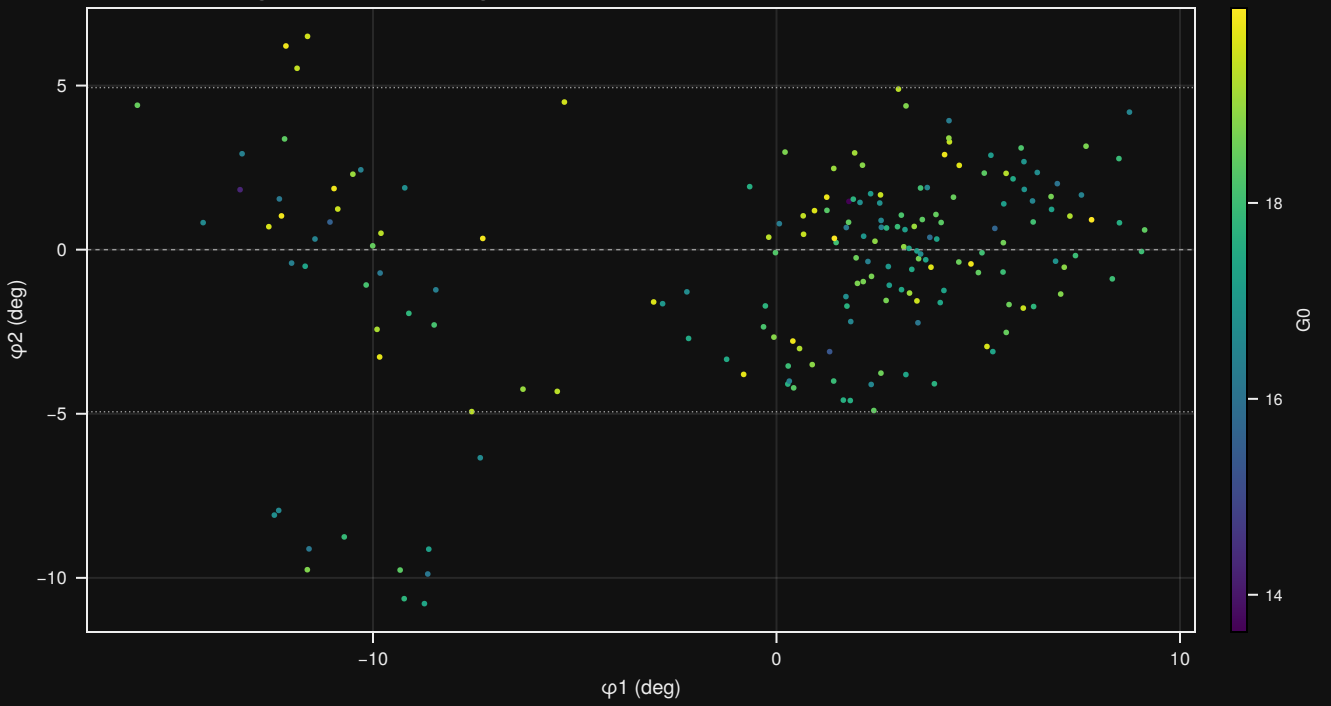
C-9

MEMBERSHIP	
Stars in catalog	183
with Gaia GSP-Phot	136 (74%)
with radial velocity	35
Stream id (sID)	35
GEOMETRY · great-circle fit	
Centroid RA, Dec	157.4, 17.4 deg
Centroid l, b	221.3, 57.8 deg
Pole RA, Dec	281.2, 58.6 deg
Length (φ1, 2–98%)	21.3 deg
Width σ(φ2), robust	2.47 deg
KINEMATICS	
pmRA* med ± σ	-8.70 ± 0.55 mas/yr
pmDE med ± σ	-5.43 ± 0.76 mas/yr
μφ1* med ± σ	-5.44 ± 0.56 mas/yr
μφ2 med ± σ	-8.80 ± 0.66 mas/yr
Parallax (cat.) med	0.273 ± 0.166 mas
Parallax (wtd, ZP-corr.)	0.291 ± 0.012 mas
Distance	d = 3.4 ± 0.1 kpc
Physical length	1.28 kpc
RV median ± σ(MAD)	39.8 ± 8.6 km/s
RV gradient dV/dφ1	0.01 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.6–20.0 (18.0)
(BP−RP)0 median	0.88
GSP-Phot [M/H] (n=136)	-1.12 ± 0.34
GSP-Phot Teff median	5542 K
RUWE > 1.4	0.0 %
PM clusters (largest)	4 (128 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	183 | 183
n_RV catalog | paper	35 | 35
<[Fe/H]> paper	-0.72
Stellar mass (paper)	630 Msun
Pole offset vs paper	9.9 deg
Remark	Several clumps?

Sky map · catalog members (n = 183) · size ∝ brightness · ring = has RV



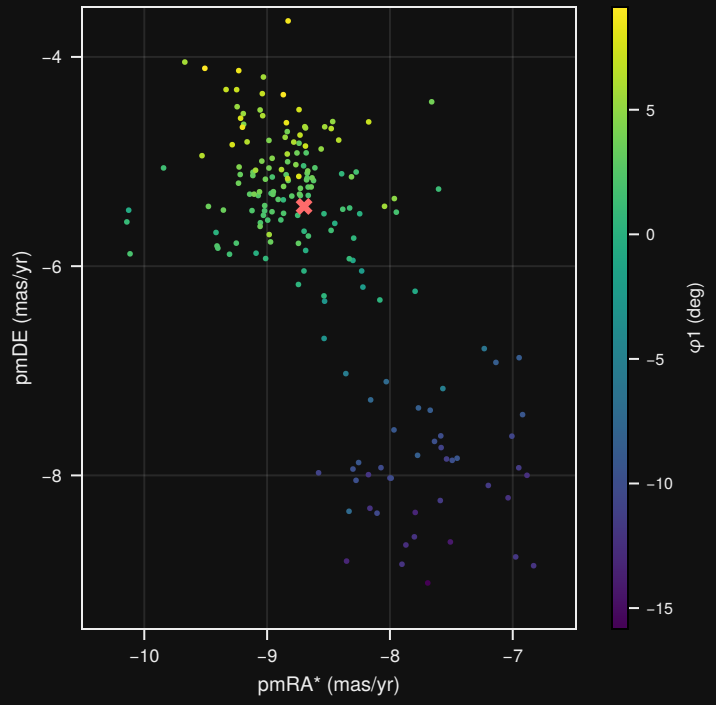
Stream frame (great-circle fit) · length 21.3° · width σ 2.47°



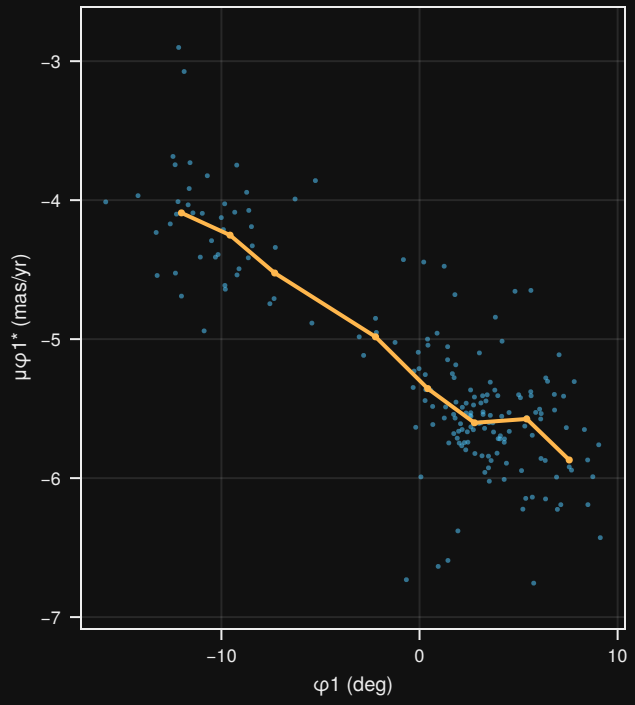
C-9

Kinematics · proper motion · parallax · radial velocity

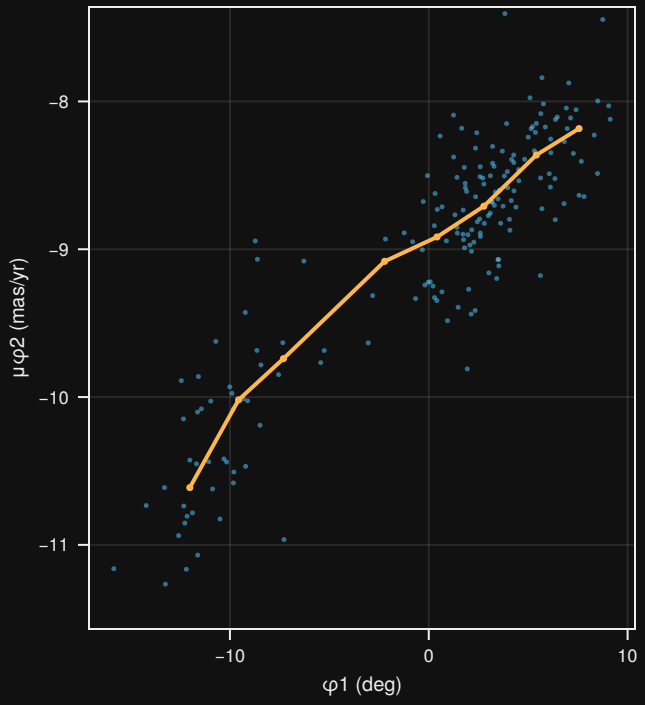
Proper motion (n = 183)



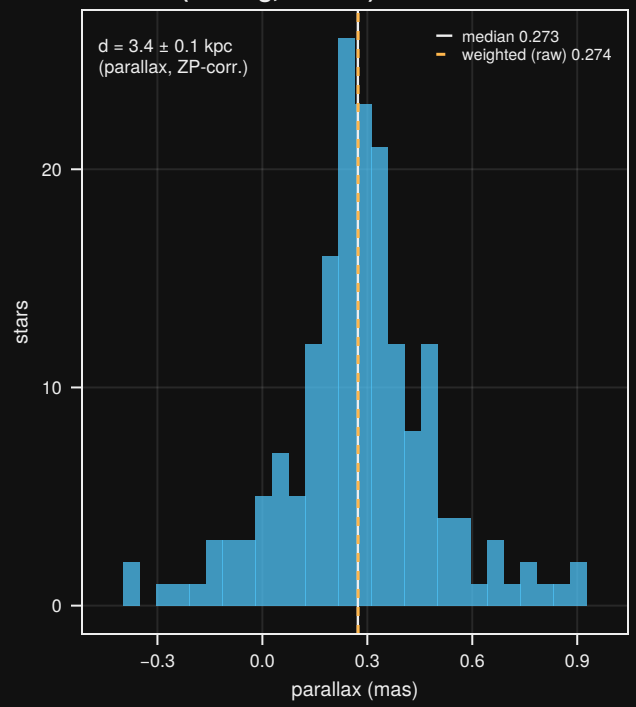
μφ1* vs φ1 · σ = 0.56



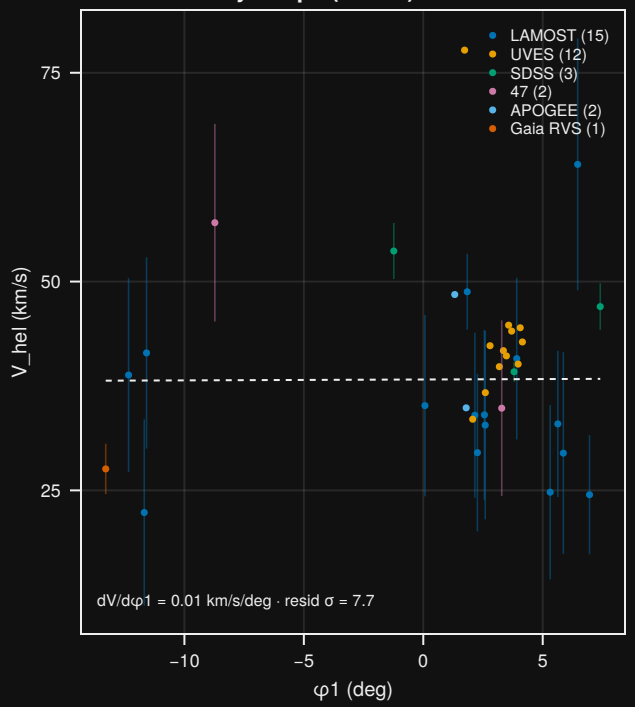
μφ2 vs φ1 · σ = 0.66



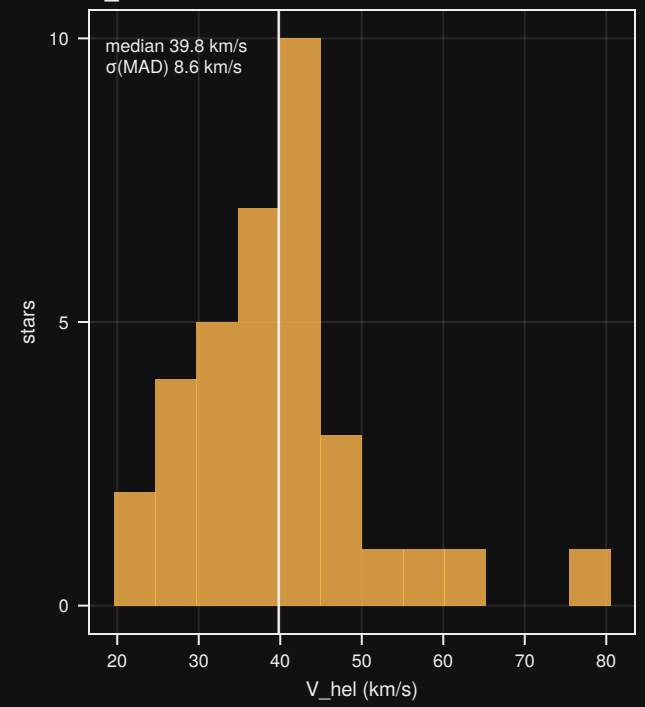
Parallax (catalog, n = 183)



Radial velocity vs φ1 (n = 35)

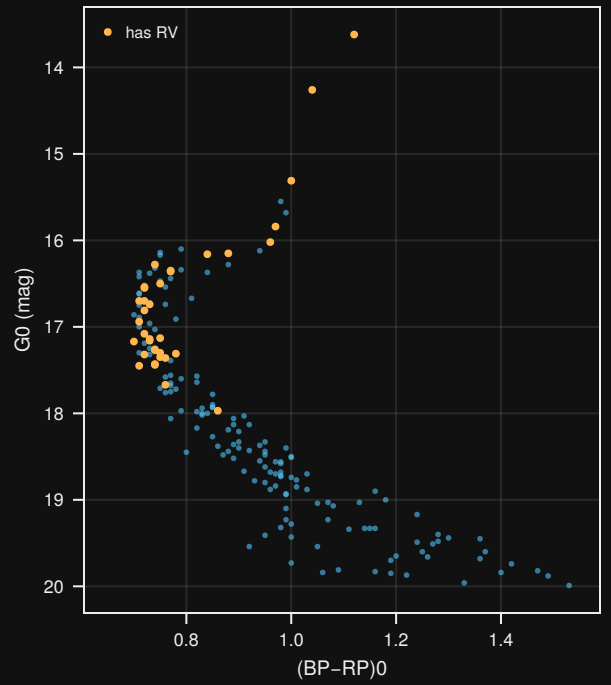


V_hel distribution

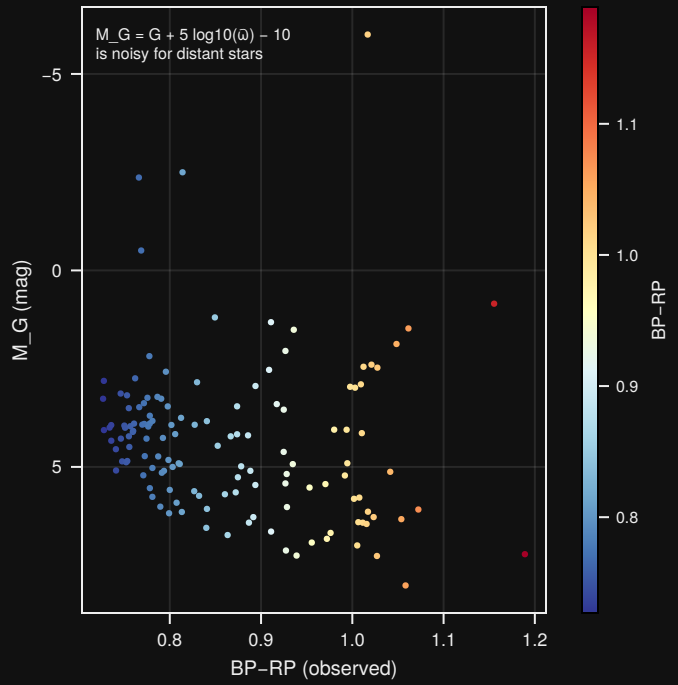


C-9

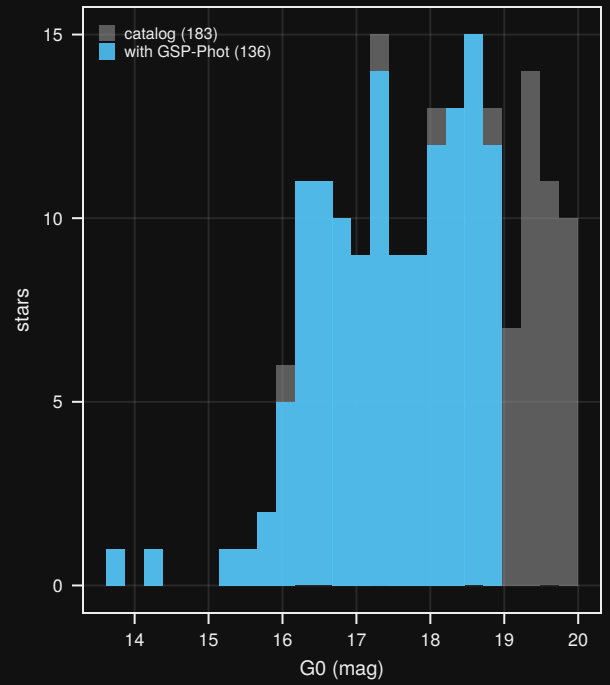
CMD · dereddened (n = 183)



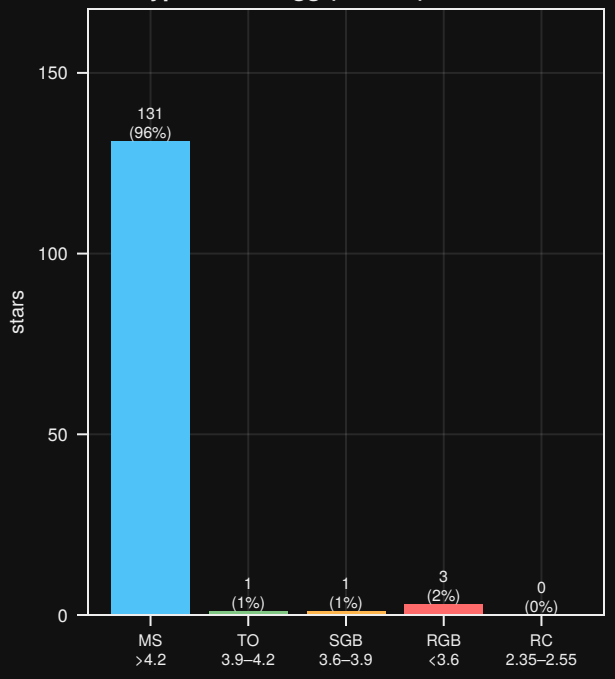
HR diagram · ϖ>0 (n = 134/136)



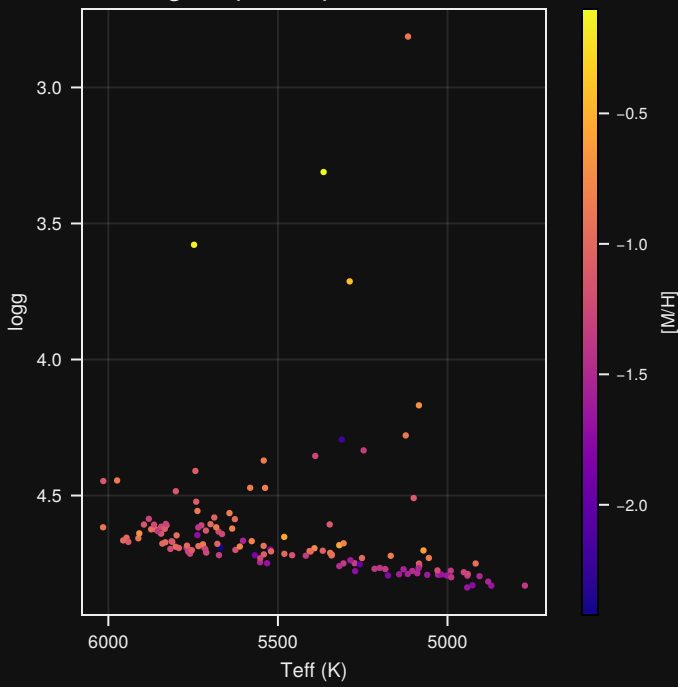
G0 & GSP-Phot coverage



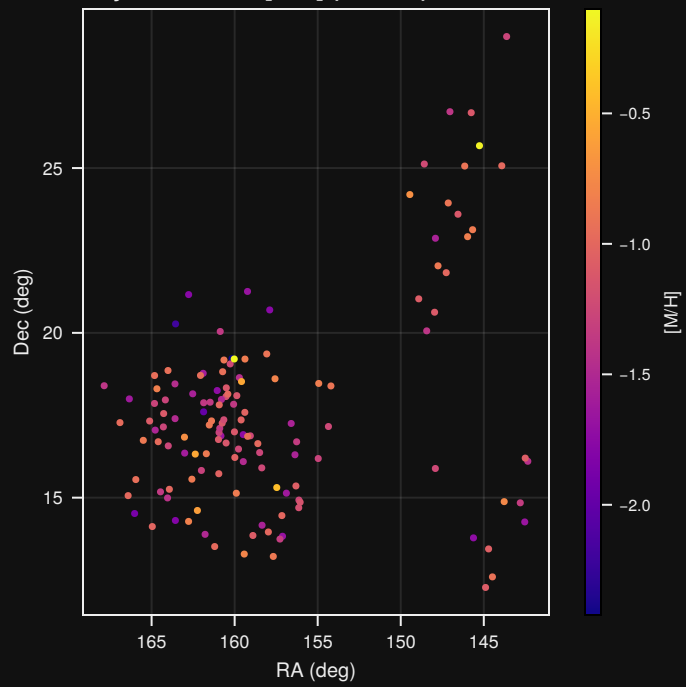
Star types from logg (n = 136)



Kiel diagram (n = 136)



Sky · GSP-Phot [M/H] (n = 136)

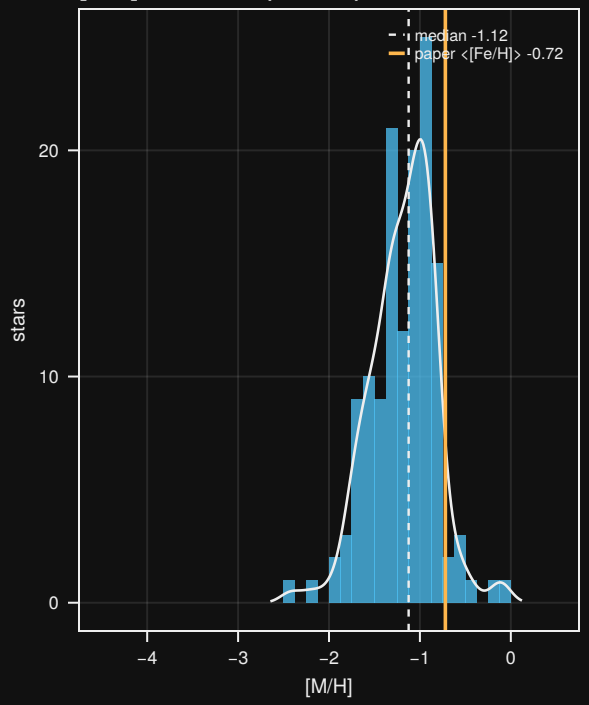


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

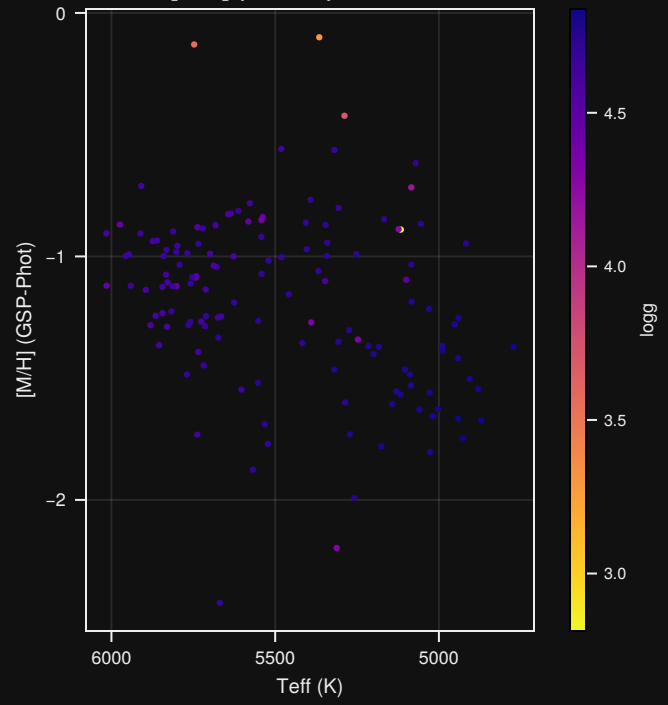
C-9

Metallicity · age · astrometric quality

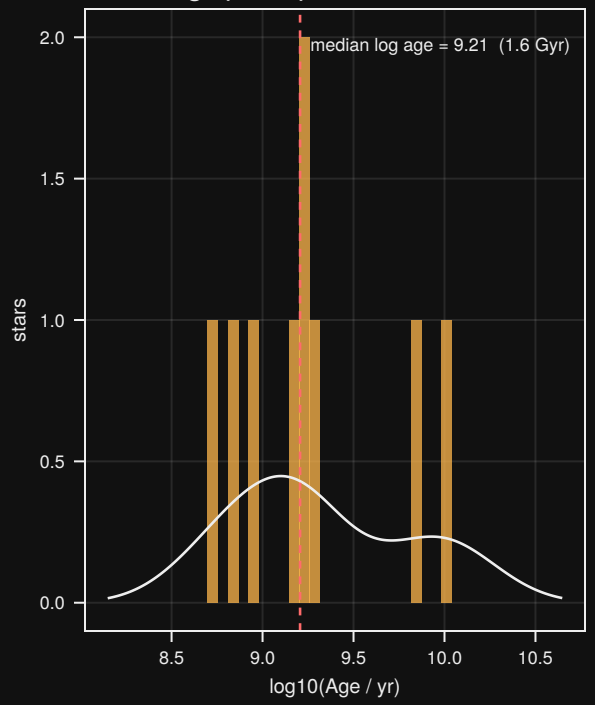
[M/H] GSP-Phot (n = 136)



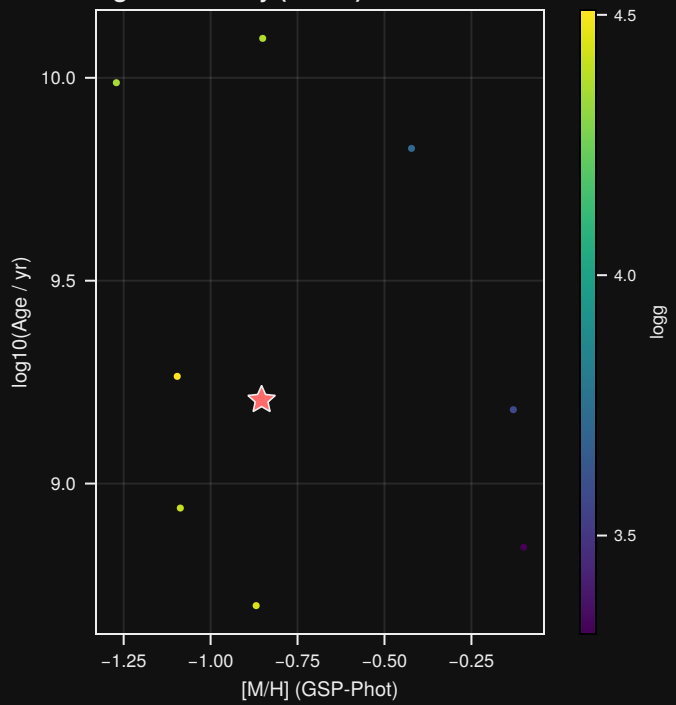
Teff vs [M/H] (n = 136)



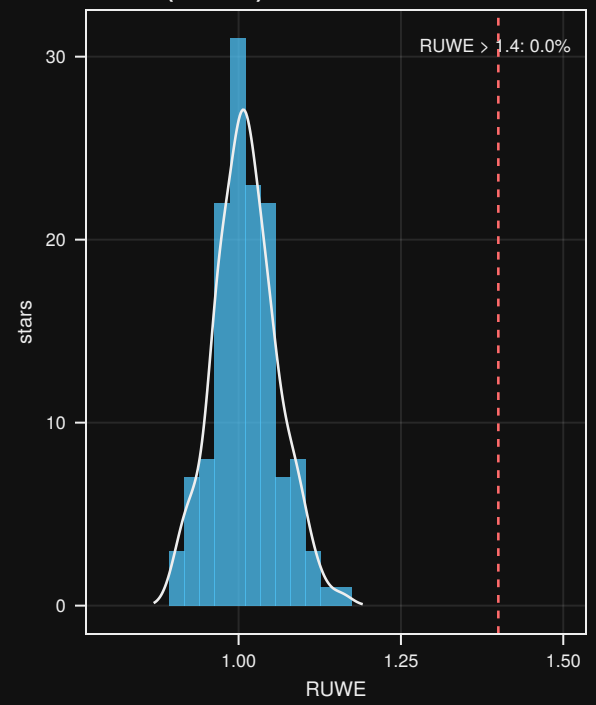
FLAME age (n = 10)



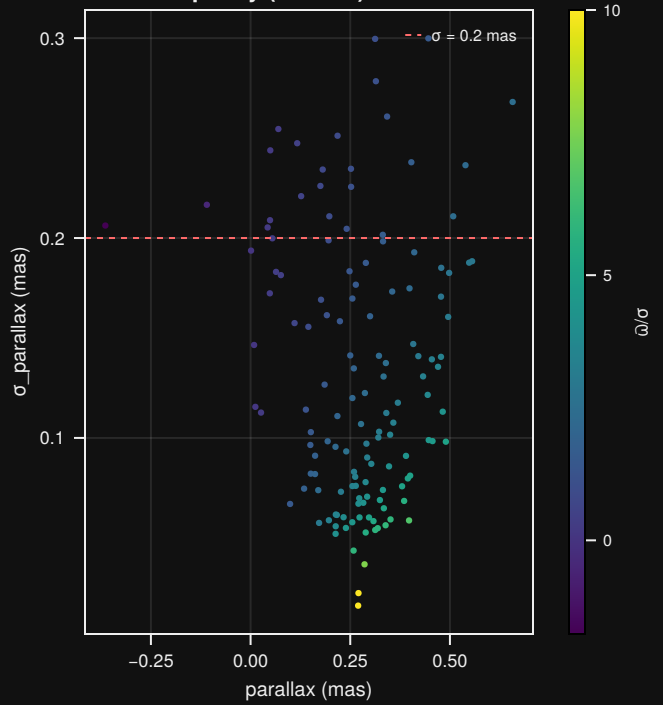
Age–metallicity (n = 10)



RUWE (n = 136)

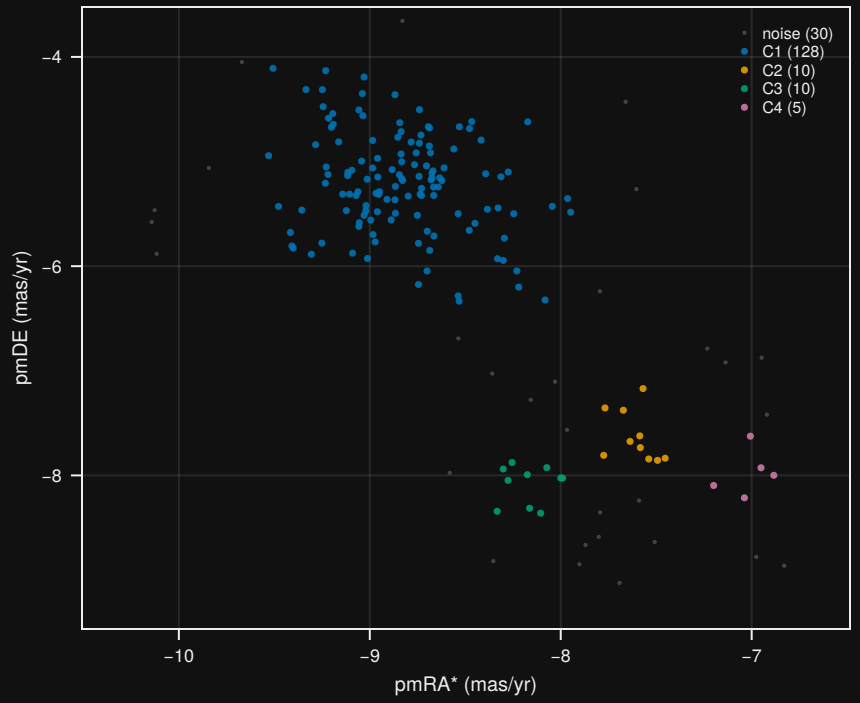


Parallax quality (n = 136)

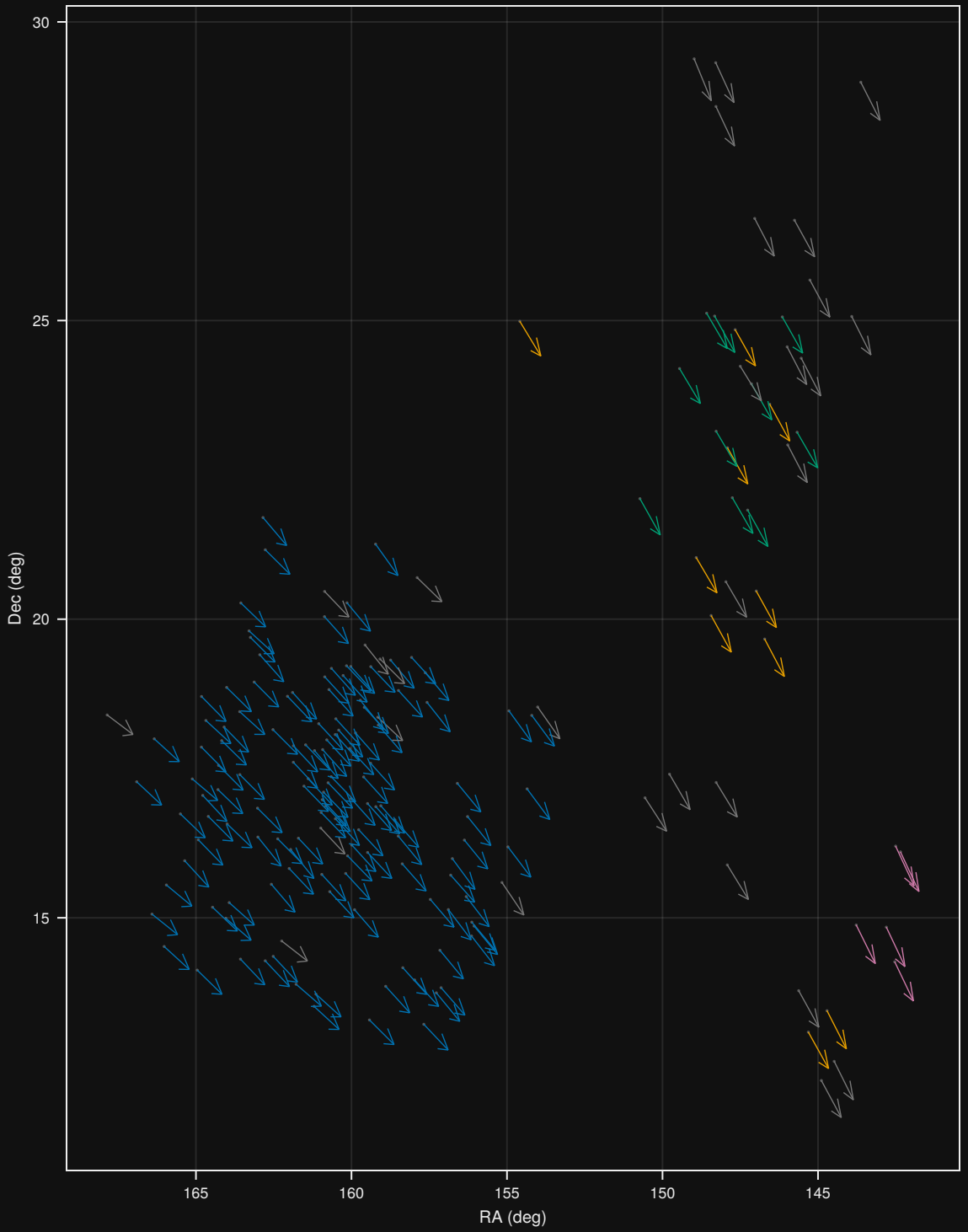


C-9

PM clustering · HDBSCAN · 4 cluster(s) · noise 16%



Proper-motion directions (arrows) coloured by cluster



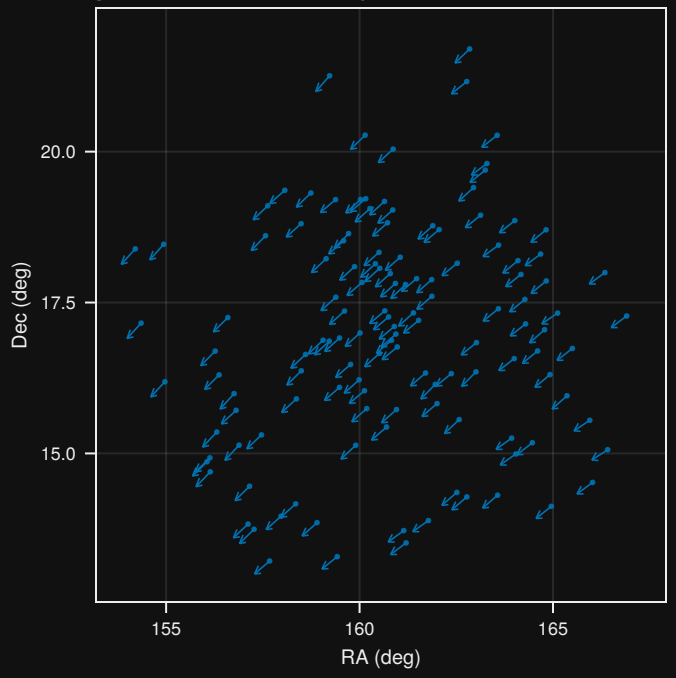
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	30	-7.84	-7.19	0.84	-1.11	3
C1	128	-8.84	-5.18	0.30	-1.16	30
C2	10	-7.58	-7.70	0.10	-1.12	0
C3	10	-8.17	-8.03	0.15	-0.89	0
C4	5	-7.01	-8.00	0.08	-1.34	2

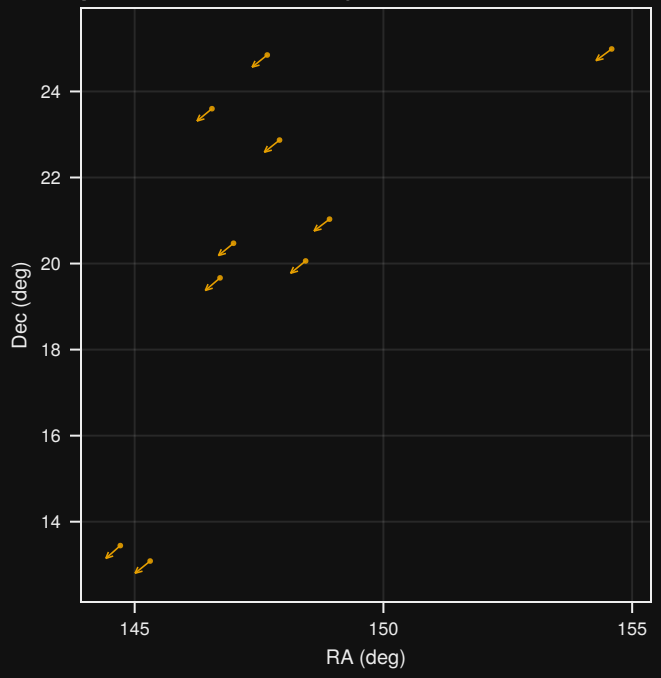
C-9

Per-cluster proper-motion vector maps

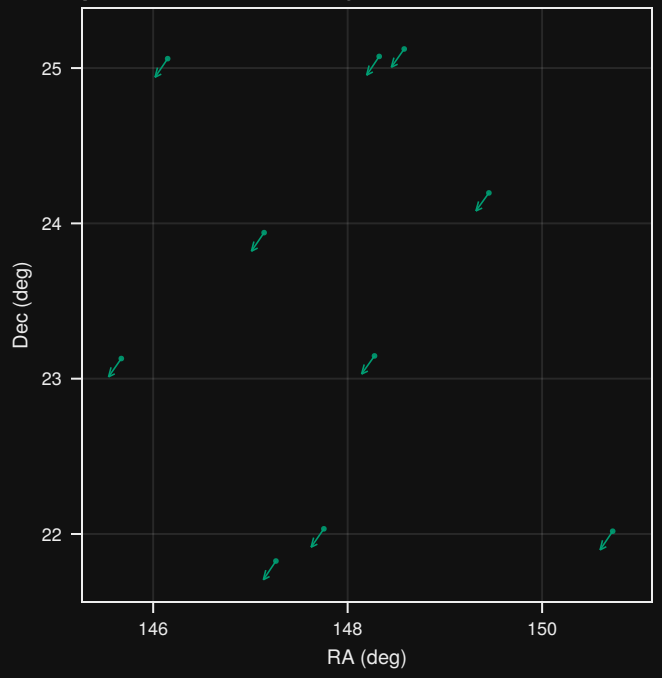
Cluster C1 · n = 128
pm = (-8.84, -5.18) mas/yr



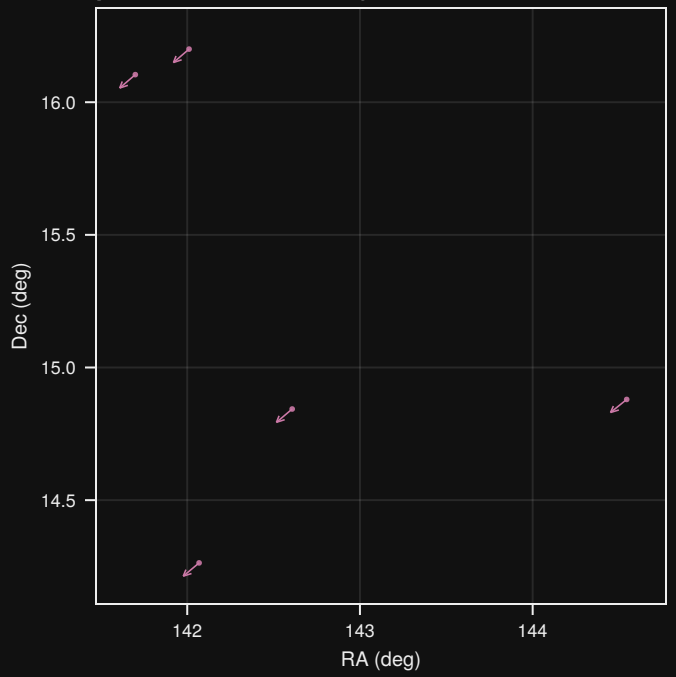
Cluster C2 · n = 10
pm = (-7.58, -7.70) mas/yr



Cluster C3 · n = 10
pm = (-8.17, -8.03) mas/yr



Cluster C4 · n = 5
pm = (-7.01, -8.00) mas/yr



Noise · n = 30

